



Supporting Online Material for
**Conditional Cooperation and Costly Monitoring Explain Success in Forest
Commons Management**

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1. Materials and methods

In this research we use field experiments, household questionnaires, and community surveys to investigate whether there is group level variation in the norm of conditional cooperation and how it affects outcomes of forest commons management, controlling for the structural determinants discussed in section 2.3 of this document. Because in groups with heterogeneous behavioral types, conditional cooperators are unlikely to sustain cooperation on their own, we also examine the scope of costly monitoring as a mechanism in coordinating the expectations of conditional cooperators in forest commons management. Our data comes from 679 members from 49 small-scale herding groups of the Bale Oromo people who were given formal rights to manage their forests as commons under a major participatory forest management program in Ethiopia. These groups are located in Adaba (17 groups) and Dodola (32 groups) districts and share social, economic, and cultural backgrounds. Here, we describe the importance of our study in relation to previous experimental as well as field studies with commons users and other small-scale societies, highlight our methodology to overcome challenges of conducting field experiments, and offer detailed analysis to support arguments in the main paper.

1.1 Importance of the study

Several key features distinguish our work from the existing studies. First, we investigate not only cooperation and monitoring patterns across different groups but we also relate these patterns to explain variation in cooperative outcomes outside the laboratory, which is forest commons management in our case. In this regard, we fill an important gap in the literature by complementing lab findings on the importance of conditional cooperation and costly enforcement (SI).

Second, we show that costly monitoring provides a mechanism through which conditional cooperators enforce cooperation. Our investigations reveal that experimental measures of conditional cooperation co-vary positively and strongly with survey measures on costly monitoring, even when we control for structural determinants of costly monitoring. This means that conditional cooperators not only contribute more to the first-

order public good, but are also more willing to contribute to the second-order public good, i.e., to enforce first-order cooperation. Our study thus advances the frontiers of previous findings that tentatively establish the importance of monitoring in sustaining commons (S2, S3) as well as studies highlighting co-variation of cooperation and enforcement in small-scale societies (S4). Moreover, by linking experimental measures of conditional cooperation with survey measures on costly monitoring we exploit further complementarities.

Third, our approach of using conditional cooperation alongside structural factors to explain variation in forest management outcomes as well as costly monitoring sets our study apart from previous studies on commons management, which are restricted to testing the importance of structural factors alone (S3, S5-8). We therefore advance the knowledge required for formulating a coherent behavioral theory of collective action and for designing policies targeting successful commons management (S9, S10).

Fourth, the participants in our study comprise non-student subjects naturally engaged in the management of forest commons. The experiments we employ emulate cooperation dilemmas that these participants face in their natural environment. Moreover, we engage a large number of participants (N= 679) representing 53 % of the member population in the 49 groups. These features allow us to measure domain specific behavior and draw reliable inferences. A related study (S11) measures voluntary cooperation among non-student subjects but they do not examine the effect of this on the outcomes of commons management.

Fifth, though we conduct this study with members from 49 diverse commons users groups, the groups strongly share social, economic, and cultural backgrounds. Thus, in comparison to previous studies which were conducted with socially, economically, and culturally diverse groups spanning many continents (S4, S12, S13), the methodological challenges we face in comparing experimental results across different groups are assuaged.

Finally, previous experimental field studies measure cooperation and punishment behaviors using the ultimatum, dictator, and third party punishment in the dictator game. Here we use the public goods game to measure behavioral heterogeneity, which is considered as a direct test of conditional cooperation (S14).

1.2 Research methodology

Our overall research strategy centers around several key aspects: One of the key issues in conducting large-scale experimental studies is to ensure that experimental outcomes are comparable across different groups and not confounded with other factors, such as sample size, contagion, contamination, experimenter effects, degree of anonymity, and payoff related feedbacks. We followed the pioneering approach to conducting experiments with small-scale societies by (S4) and took several steps to minimize these effects. These are discussed below:

Reliable measures of cooperation outcome in the field

Our research strategy was to select a study area with a high-quality and reliable measure of forest management outcomes. In the context of our study area, forest mensuration data on young trees provides not only a context specific but also a statistically reliable bio-physical measure of cooperation. The data on young trees is collected for each group by trained personnel from the forest administration with active participation by group members (S15). It was provided to us by the program office. We give detailed information on the forest management outcome in section 2.1 of this document. (S16) offers a more comprehensive review.

Number of groups (observations)

Our aim was to run a quantitative analysis using a combination of statistical tests and econometric models. We therefore chose a study area with a sufficiently high number of group level observations to run such an analysis. Our sample size of 49 forest user groups is in line with previous econometric studies on common property management (S5-8).

Robust classification of behavioral types

Measuring conditional cooperation in the field encounters many difficulties. For instance, it is not possible to infer the absence of conditional cooperators from a group in which little cooperation is observed. If conditional cooperators believe that no one else will cooperate

they will also not cooperate. To circumvent this problem, we follow the experimental design by (S17), in which we control for a subject's belief about cooperation by other group members and thus provide an explicit measure of conditional cooperation. Further, this method has been replicated by (S18, S19) and found to yield reliable results.

Sample size

We aimed at recruiting participants who are highly representative of the member population. Our strategy to achieve this was to employ a large sample. Our experimental and household questionnaire data come from 679 and 459 members, representing 53 % and 36 % of all members in the 49 groups, respectively. All individuals who answered the household questionnaire also participated in the experiment.

Written instructions

We used fixed written instructions (section 1.5 below), set of examples, and control test questions as our guidelines for conducting the experiments. This not only ensured uniformity in procedures and verbal explanations at all groups but also that subjects understood the games well enough to participate in them. The instructions were first tested in a pilot study in a different region of Ethiopia, which is linguistically similar to the Bale region. Based on the feedback, the instructions were further fine-tuned and tested again in three forest user groups of the Denaba village (Birbisa Guxa, Bulchana Hubo, and Edo Wixe). These groups were excluded from the main study and from all results discussed in the paper. Our protocol also prevented people from discussing the game among each other.

Game endowment and actual currency

The participants in our study frequently engage in monetary transactions in local markets, where they buy and sell forest, animal, and agricultural products. As a result they are well familiar with money, especially small currency bills. Bearing this in mind, we used actual bills of one Ethiopian Birr while delivering instructions, examples, testing, and for playing the actual game. In addition to the familiarity with money, using one Birr denomination

also made it easier for the participants to make simple calculations. Further, we standardized the game stake at one-day's wage in the main town (Dodola) of the study area.

Contagion

In our field setting, contagion and contamination are possible when forest user groups border each other or are located within the same village. We controlled for this by adopting a two-fold approach: (i) focusing on groups from one village at a time, and (ii) inviting, if possible, two neighboring groups to play the games on an appointed date. In case two groups were invited, we did not ask one group to travel to the other group. Rather, we chose a regular meeting point for members from many different groups in a particular region. Experiments began a day after the invitations were sent and were run each day, without a break, till all groups from a village had taken part in all the experiments. Once this was achieved, we moved to the next village and followed similar procedures there. Because each group was engaged in experiments that lasted an entire day, conducting experiments among neighboring groups in a village meant that members who took part in the experiments were left with less time to interact with members from the groups yet to take part in the experiments.

Overall, in 19 groups, experiments were conducted separately, whereas in the remaining 30 groups, two groups were invited at a time to take part in the experiments. We test for potential effect of this on behavioral type composition but find no statistical difference in the share of conditional cooperators (Mann-Whitney $z = 0.32$, $P = 0.750$) or free riders (Mann-Whitney $z = -0.82$, $P = 0.410$) or other types (Mann-Whitney $z = -0.28$, $P = 0.782$) across the two situations.

Experimenter effects

All experiments were run by the main author together with the main assistant. Further, all interviews were conducted by the same two field assistants. To minimize experimenter effects further, we used uniform instructions in all the 49 groups. When two groups were invited at the same time, one group was guided through the experiment by the main author, the other group by the main assistant. We do not find any evidence of an experimenter

effect on either the share of conditional cooperators (Mann-Whitney $z = -0.27$, $P = 0.789$) or free riders (Mann-Whitney $z = 0.47$, $P = 0.637$).

Anonymity

Participants were ensured that their decisions will be anonymous to each other as well as to the two field assistants. Moreover, even though the main assistant is a native speaker, he is not from the study area and hence was not personally known to any of the participants. Further, like experiments, household interviews were done in private and responses were anonymous to other participants.

Payoffs

The society members took part in three experiments, of which the experiment discussed here, was the first. We paid subjects their earnings from all the games in the end.

1.3 PFM setting, subject pool, and socio-economic background

The participatory forest management (PFM) program is operational in Dodola and Adaba districts (Woredas) of the Bale Zone, Oromia State. Both districts are bordered on the west, east and south by other districts of the Bale Zone. To the north flows the Shebelle River, beyond which lies the Arsi Zone of the Oromia State. Thus, the entire program area falls within the Bale Zone and does not border non-Oromo groups either from Somalia or the Somali state of Ethiopia, the Ogaden.

The participants in our study comprise 32 forest user groups from three villages of Dodola district and 17 forest user groups from two villages of Adaba district. Here, we would like to draw the attention to the difference between a forest user group and a village. A forest user group is a small settlement within a village, such that a village comprises many forest user groups plus other groups that are not part of the PFM program (we refer to these as non-PFM groups). Within each village, forest user groups are typically located in the mountains, whereas the non-PFM groups are located in the plains. Below, we present a Google image of Denaba village from Dodola district (Fig. S1). In the image, forest user groups are located in the mountains and the forest blocks managed by these groups are

located behind them, but the non-PFM group is located in the plains (centre of the image), around 4 km away from the nearest forest user group in our study. Dodola, the main town is located on the left, and is bisected by the main road, which is the only road in our study area; it connects the towns of Dodola and Adaba.

Forest user groups are primarily responsible for the management of their forest blocks and the village administration's role is at the most in the settlement of disputes between forest user groups and non-PFM groups. The model bylaw for forest management (S20) to which all forest user groups are signatories, clearly states that

“the highest decision making body of the WAJIB¹ is the general assembly that includes all recognized members. Relevant bodies such as representatives of village administration and district administration may participate in WAJIB general assembly meetings by invitation. In the event of a conflict between members, the executive committee and the general assembly shall take the necessary measures to resolve it. A higher level involvement (village administration or court) may be sought if the conflict cannot be resolved by them”.

Due to the remote location of the forest user groups, the interaction between members from these groups and the main village settlement is restricted to either market days (once a week) or special occasions, such as marriage, funerals etc.

¹ WAJIB stands for Waldayaa Jiraatoota Bosonaa in Afaan Oromo and means “forest user group”.

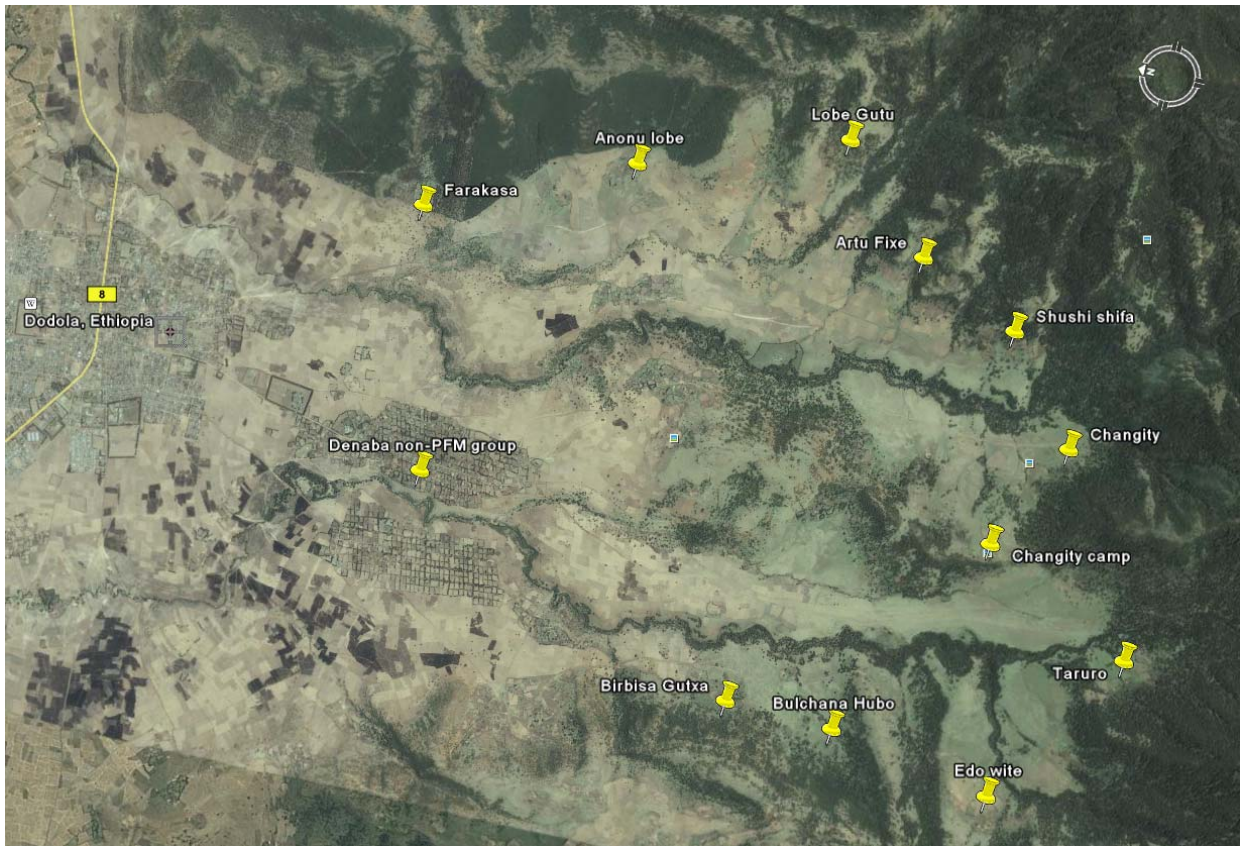


Fig. S1. Google image showing the location of forest user groups in the Denaba village, Dodola district.

In Table S1 we summarize the main socio-economic variables of our subject pool from different forest user groups. The data on these variables was collected through household questionnaire. Group level data on these variables which we use in the regression analysis is summarized in Table S8.

Variable	Mean	SD	N
Female	10.02 %	-	619
Religion: Islam	99.03 %	-	619
Age (years)	36.75	13.607	616
Education (years)	3.44	3.104	615
Family size	8.59	4.327	604
Number of cattle	11.51	10.505	611
Number of horses	2.66	2.072	612
Land holding (hectares)	1.51	1.121	613

Table S1. Socio-economic background of the subject pool.

1.4 Experimental procedures

Participation in experiments was voluntary and informed consent was taken from all participants before the experiments began. Here we enlist in detail the typical procedures we adopted at each stage of the experiment.

Pre-game preparation

We employed community mobilizers, one from each village, to organize appointments with the groups on our behalf. The community mobilizers knew nothing about the experiments. They were informed: (a) to invite group members to arrive at an appointed date to play some games in which they could earn some money, (b) participation in the games was entirely voluntary, (c) the experiments could last an entire day.

The mobilizers were given strict instructions to invite 15-25 persons registered as members from each group. In addition, group leader, vice leader, and group committee members were highly encouraged to take part in the experiments. In some cases, members could not show up and sent a representative from their household. Representatives above 16 years old were allowed to take part in the experiments. Most experiments took place at administrative offices and tourist camps.

Game preparation

Once enough participants had arrived, we invited them to a room and requested them to sit according to their group affiliation. Participants were not allowed to enter the room if they were late by 15 or more minutes. Two local field assistants made sure that latecomers were sent back with a show up fee. After this, we requested the leaders or vice leader to check if: (i) all participants were from the invited group/s, (ii) did not belong to the same household, and (iii) were at least 16 years old. Participants who did not meet these criteria were sent back. In addition, nursing mothers with babies, sick participants, very old, and those with other health problems were also sent back with a show up fee.

Experimental instructions

Before the experiments began, the main author and the main assistant introduced themselves in the local language, *Afaan Oromo*. After this, the main assistant took over the task of reading out the written instructions to the entire group. The instructions included the following key points: (a) all game decisions are anonymous, except to the main author and the main assistant, (b) all games will be played only once, (c) separate instructions will be given for separate games, (d) control questions will be used to test game comprehension and only those who answer these questions correctly will be allowed to participate, (e) once the actual game has begun, game discussion is not allowed and will lead to expulsion from the game, (f) questions concerning games are encouraged, and (g) in case two groups are invited, members will play only with members from their own groups.

Because most of our subjects had limited literacy, we designed instructions in a didactical style. During the instructions, the participants were given lots of opportunities to ask questions. We allowed asking questions aloud because in the pilot study we observed that most participants had similar questions, so answering these questions at once saved time. Further, we realized during the pilot study that this approach helped in developing a wider understanding among participants, including those who had doubts but were hesitant to ask. We did not witness any players speaking about their strategies aloud. Most questions were either related to repeating the key steps in the game or the payoff calculation. Any unexpected question, for which we did not have a script, was answered

first by the main author and then translated by the main assistant. The assistant was not allowed to answer any question on his own.

Instructions were followed by a fixed set of examples, which were illustrated by the main author. This had many advantages: (a) participants got used to the main author speaking *Afaan Oromo*. We consider this important because when two groups were invited, the main author took over the responsibility of testing and engaging members from one of the two invited groups in the game; (ii) it gave some respite to the main assistant; and (iii) the main author gained participants' trust. The main assistant made sure that participants understood the main author's accent; none of the participants complained about this. Moreover, the participants were very happy that the main author could speak their language.

Before the actual game, the participants were tested one by one for their game comprehension using a fixed set of control questions. These questions focused on testing a participant's basic understanding of the game, such as addition, multiplication, division, and other skills. While the main assistant tested this for members from one group, the main author did so simultaneously for members from another group. All participants rejected by the main author were tested again by the main assistant. Those who could not answer the questions correctly were given a show up fee and sent back. Further, two local assistants made sure that the rejected participants and uninvited persons had left the experimental venue.

The actual game

Depending on their group, the selected participants were given plastic identity cards bearing names of Swiss Cantons and German States. Before the actual game began, the selected participants were given another opportunity to ask further questions. During the actual game, participants were called one by one to a room. Inside the room, subjects were given six bills of one Birr and then asked to make an unconditional decision to contribute to the public good. Once all participants had taken part in this decision, they were called one by one again to take conditional decisions for all possible contribution levels of the partner player. After each decision, the players left the room and sat outside, but separately from

those who were waiting to play the game. In the meanwhile two local assistants conducted household interviews and made sure that no one discussed the game. As in the experiments, anonymity was also ensured during the interviews. We were able to match a participant's behavior in the game with the interview through his/ her experimental identity (Swiss Cantons or German States).

Payoffs

To ensure incentive compatibility, both the unconditional and the conditional decisions were made payoff relevant. A dice was rolled to determine for which player the unconditional decision is taken; this was matched with the partner player's conditional decision to calculate the payoffs.

1.5 Experimental instructions

The specific instructions we used in our experiments were the following.

Introduction

Greetings and welcome to all of you. My name is Devesh Rustagi and I am a student from a university in Switzerland. I am here for a research concerning livelihood improvement through forest conservation. For my research, I would like to play a few games with you. Depending on the decisions made by you and other players in these games, you can win some money. The payment that you receive from these games is not from my own pocket, but sponsored by the German government. Before we proceed with the games, I would like to tell you some important things.

General instructions

1. In all games, your identity will be kept anonymous. This means that except for me and my assistant, no one will come to know of your identity. I am interested only in the decisions made by you in these games and not your identity. This is the reason that we will not ask your name in any of the games. We will identify your decision in the game with an identity card like this (*show plastic cards*). Please do not lose this card.

2. All games will be played for one round only. This means that after you have played the first game, we will begin with the second game. Likewise, after the second game is over, we will begin with the third game. This means that after a game is over, there is no subsequent interaction.

3. We will play three games with you, but you will receive your money only in the end. We will keep a record of your earnings in all the games on a sheet like this (*show payoff sheets for clarity*) to make sure that you receive the correct amount.

4. We will give you separate instructions and examples on how to play each of these games. The instructions for each game will be given before we play the game. For instance, before we play the first game, we will give you the instructions on how to play the first game. Likewise, when we play the second game, we will give you the instructions for the second game, and so on. It is very important that you listen to these instructions carefully. In case you do not understand the game, please stop us and ask us. We will be happy to help you.

5. Before we play the actual game, we will check if you have understood the game or not. In case you do not understand the game, we will give the instructions again. However, if you are still not able to understand the game, we will have no choice but to request you to leave the venue. In this case, you will receive five Birr from us. Therefore, it is important that you listen to the instructions carefully.

6. We would like to keep the game anonymous, therefore, please do not discuss the game with each other. But you may discuss about politics, rainfall, market, cattle, WAJIBS, and other such things. In case we find that you are discussing the game with other players, we will exclude you immediately from the game. In this case, you will not receive any money.

7. We also request you to not to discuss the game with other WAJIB members as this will spoil my study.

8. (*Read this only when two societies are invited*) You will play the games only with the members of your own society.

9. I repeat again, please do not hesitate to ask any questions. We encourage you to ask as many questions concerning the games as possible. In case you have any questions at this stage, you may ask them now. Otherwise, we will begin with the instructions for the first game.

Experimental instructions

We will now give you instructions and examples for the first game. There are two parts in this game. We will now give you instructions for the first part. This is followed by a test in which we will check if you have understood the game or not. Once we are sure that you have understood the game, we will begin playing the game.

- In this game, we will divide you into groups of two players. You will not come to know to which group you belong. Likewise, you will not come to know the identity of the other (partner) player in your group. Similarly, the other player will not come to know your identity.
- At the beginning of the game, each player will receive six Birr from us. Now you have to decide how many from the six Birr to put into your pocket and how many into a project. You may put any amount between 0 and 6 Birr into the project.

Now we will show you how this is done. Please note that since this is an example, we will tell the player how many Birr to put into the project. But when we play the actual game, you will have to decide this on your own, without any help from us. *(Randomly select a player and give him six bills of one Birr each. Please make sure that each time YOU tell the person on how much he should put into the project. Do not allow the player to take a decision because this may influence the decision of other potential players).* Suppose you are a player in this game. As mentioned before, you receive an endowment of six Birr from us. Now let us assume that out of six Birr, you put zero Birr into the project. Please put zero birr into the project. *Ask the group:* Can you tell me how many Birr there are in the project? How many Birr does the player have in his pocket? Have you understood this?

Now, let us assume that out of six, you put one Birr into the project. Please put one Birr. How many Birr are in the project? How many Birr does the player have in his pocket? (*Carry on this procedure till 6 Ethiopian Birr*).

Have you understood this part? Do you need additional examples? (*If yes, select another person and repeat the examples in the same order*).

- Any amount in the project will be increased by the same number of Shillingis² as the number of Birr in the project. For example, if you put 0 Birr into the project, the project amount will be increased by 0 Shillingis. Now, the final amount of money in the project is 0 Birr. If you put 1 Birr into the project, the project amount will be increased by 1 Shillingi. Now, the final amount of money in the project is 1.5 Birr (*Carry on till 6 Birr*). I repeat, the project amount will be increased by the same number of Shillingis as the number of Birr in the project. Have you understood this? Do you need additional examples? (*If yes, select another person and repeat the examples in the same order*).
- After the project money has increased, it will be divided equally between you and your partner player, irrespective of how much you have put into the project (*Please repeat this again*). For example, if the project contains 0 Birr, it will be increased by 0 Birr and then divided equally between you and your partner player. However, since zero does not increase, both you and your partner will get zero Birr from the project. For example, if the project contains 1 Birr, it will be increased by 1 Shillingi. Now the total value of the project is 1.5 Birr, and both you and your partner player get 0.75 Birr each from the project (*carry on till 6 Birr*). Have you understood this part? Do you need additional examples? (*If yes, select another person and repeat the examples in the same order*).

Please remember that any money that you put into the project is first increased and then divided equally among the players in your group. Any amount that you put in your pocket remains the same. If you put 1 Birr in your pocket, it remains 1 Birr. It neither increases nor is it divided.

² 1 Shillingi = 0.5 Ethiopian Birr

- Your final earning from the game is the sum of the amount you have in your pocket and the amount you receive from the project.
- We will now give you three examples. Please note that since now we are learning how to play this game, you can see the identity of each player as well as the decisions made by them. When we play the actual game, you will not come to know of this. Do you understand this? We will now select two people and tell them to take the following decisions in the game. You are player I and you are player II (*look for participants with weak comprehension and always give them a chance to act as player I and II*). We give you 6 Birr each at the start of the game.

Example 1: Now we will see what happens if both players put zero Birr into the project. Player I and II: Please put zero Birr into the project. Now, can you tell me how many Birr did player I put into the project? How many Birr does he have in his pocket? How many Birr did player II put into the project? How many Birr does he have in his pocket? How many Birr are in the project? We have zero Birr in the project. Since zero Birr does not increase and cannot be divided, each player gets zero Birr back from the project.

Player I has put zero Birr into the project, so he has six Birr in his pocket. He gets zero Birr from the project. Can you tell me, what is his income? Since player I has six Birr in his pocket and he gets zero Birr from the project, his final income is six Birr. (*Please repeat the procedure to calculate the income of the second player.*)

Example 2: Now we will show you the second example. You are player I and you are player II. You get six Birr from us at the beginning of the game. Now we will see what happens if both players put six Birr into the project. Player I and II please put six Birr into the project. Now, can you tell me how many Birr did player I put into the project? How many Birr does he have in his pocket? How many Birr did player II put into the project? How many Birr does he have in his pocket? How many Birr are in the project?

We have 12 Birr in the project. The project amount will now be increased by 12 Shillings. The final amount in the project is $12 \text{ Birr} + 12 \text{ Shillings} = 18 \text{ Birr}$. Now 18 Birr is divided equally among both the players. So, each player gets 9 Birr.

Now, can you tell me, how many Birr does player I have in his pocket? How many Birr does he get from the project? What is his final income? We repeat, since player I has zero Birr in his pocket and he gets nine Birr from the project, his final income is nine Birr. (*Please repeat the procedure to calculate the income of the second player.*)

Example 3: Now we will show you the third example. You are player I and you are player II. We will see what happens if player I puts zero Birr into the project and Player II puts six Birr into the project. Player I, please put zero Birr into the project and Player II, please put six Birr into the project. Now can you tell me how many Birr did player I put into the project? How many Birr does he have in his pocket? How many Birr did player II put into the project? How many Birr does he have in his pocket? How many Birr are in the project?

We have six Birr in the project. The project amount will be increased by 6 Shillings. So the final amount in the project is $6 \text{ Birr} + 6 \text{ Shillings} = 9 \text{ Birr}$. Now 9 Birr is divided equally among both the players. So, each player gets 4.5 Birr. Now, how many Birr does player I have in his pocket? How many Birr does he get from the project? So, what is his final income? We repeat, since player I has 6 Birr in his pocket and he gets 4.5 Birr from the project, his final income is 10.5 Birr. How many Birr did player II put into the project? How many Birr does he get from the project? So, what is his final income? I repeat, since player II has zero Birr in his pocket and he gets 4.5 Birr from the project, his final income is 4.5 Birr.

- We will now summarize the key results from these examples:
- a) If both players put zero Birr into the project, they both earn 6 Birr.
 - b) If both players put 6 Birr into the project, they both earn 9 Birr.
 - c) If one player puts zero and the other player puts six Birr into the project, the player who puts zero Birr earns 10.5 Birr, while the player who puts 6 Birr, earns 4.5 Birr.

- d) If you and your partner player put the same amount into the project, you both earn the same income.
 - e) If you put less than what your partner puts into the project, you earn a higher income.
 - f) If you put more into the project than your partner, you earn a lower income.
- If you have any questions, you may ask them now. Otherwise, we will call you one by one and ask six questions to check if you have understood the game or not. Please note that if you answer these questions wrong, we will give you 5 Birr and request you to leave the game venue. Therefore, please tell us if we need to repeat the examples or not (*If yes, repeat the examples in the same order*).

Control questions

1. How much money do you get at the start of the game? / What decision do you have to take in the game?
2. Suppose, you decide to put X^3 Birr into the project, how much is left in your pocket?
3. What happens to the money in the project?
4. If you put X Birr into the project, by how much will this increase? What happens after the money is increased?
5. If you put X Birr into the project and your partner also puts X Birr into the project, who earns more?
6. If you put Y Birr into the project and your partner puts Z Birr into the project, who earns more?

(For those who answer 5-6 questions correctly, ask them to sit back in the room. Pay the remaining players 5 Birr and request them to leave. After this, repeat the control questions and let the selected players answer in a chorus. Ask again, if everyone understands. If yes, give them the identity cards).

³ X varied from player to player.

- We will now call you one by one to enter this room and play the game. Please remember that you will not come to know the identity of your partner player or the amount they put in the project.
- We will also ask you a question: How many Birr do you believe your partner player will put into the project? This is an important question, so please think before you answer this question.
- While you wait for your turn, two assistants will conduct interviews with some of you. They will also check if you discuss the game with each other or not. If they find you discussing the game, we will have to expel you from the game.
- When entering the room, please keep your identity card ready.

Unconditional decision

Hello! Have a seat please. I hope you have understood the game. Your identity card, please? Here are your six Birr. Now you have to decide out of six Birr how much you would like to put into the project. Please put the amount here on the table. How many Birr do you believe your partner player will put into the project? Thank you. Please do not discuss this with the other players.

Additional experimental instructions for the conditional decision

We will now give you instructions to play a slightly different version of the decision that you just played. In the first decision, you did not know the amount your partner player puts into the project. But in this game, we will tell you how many Birr your partner player puts into the project. After you have seen this, you can decide on how many Birr you would like to put into the project. There are seven decisions to be made in this game. Each decision is independent of the other. Please note that you will get a fresh endowment of six Birr at the start of each decision. We will now give you illustrations on how this game is played. Please listen carefully. While we give examples, no one is allowed to speak.

Decision 1: Your partner player in the game puts out of six - zero Birr into the project (*put no money on the table*). Now, out of six Birr, how much would you like to put into the project? After you have made your decision, the decision is over.

Decision 2: Your partner player in the game puts out of six - one Birr into the project (*put one Birr on the table*). Now, out of six Birr, how much would you like to put into the project? After you have made your decision, the decision is over. (*Carry on till 6 Birr.*)

Do you have any questions?

There are seven decisions to be taken in this game. Please watch our fingers for why there are seven decisions. Your partner player puts 0, how much would you like to put; Your partner player puts 1, how much would you like to put; Your partner player puts 2, how much would you like to put; Your partner player puts 3; how much would you like to put; Your partner player puts 4, how much would you like to put; Your partner player puts 5, how much would you like to put; Your partner player puts 6, how much would you like to put. Can you count our fingers now? How many decisions do you have to take in this game?

At the beginning of each decision, you will get 6 Birr, just like in the examples you saw. Each decision is independent of the other. A very important point is that we will pick only one of these seven decisions to decide your earnings. So please take all the decisions seriously. Do you have any questions?

We will now call you one by one to play this game. As usual, please keep your identity card ready.

Conditional decision

Hello! Please take a seat. I hope you have understood the game. Your identity card, please? Here are your six Birr. Now we will show you one by one how much your partner player puts into the project. After you have seen this, you can decide how many Birr you would like to put into the project. Please put the amount here on the table.

Decision 1: Your partner player in the game puts out of six - zero Birr into the project (*put no money on the table*). Now, out of six Birr, how much would you like into put in the project? Now this decision is over. Please return all the money you have in your hand to me.

Decision 2: Here are your six Birr. How many Birr do you have in your hand? Your partner player in the game puts out of six - one Birr into the project (*put one Birr on the table*). Now, out of six Birr, how much would you like to put into the project? Now this decision is over. Please return all the money you have in your hand to me.

(And so on till 6 Birr.)

1.6 Household questionnaire

8-13 members from each group were interviewed in detail using a household questionnaire. All of these members also took part in the experiment. The following topics were covered in the household questionnaire:

- a) Socio-economic and demographic profile
- b) Forest dependence: fodder, timber, non-timber products
- c) Market access
- d) Leadership
- e) Relations with non-members
- f) Relations with the local government
- g) Participation and monitoring
- h) Overall feedback on the program

Those players who could not be administered detailed household questionnaires were administered shorter ones covering only socio-economic and demographic aspects.

1.7 Community questionnaire

Each group was also administered a community level questionnaire in which 5-6 members, not interviewed earlier using the household questionnaire, were asked to respond to questions covering the following main topics:

- a) Settlement heterogeneity
- b) Clan heterogeneity
- c) Forest rent payment mechanism
- d) Relation with members from neighboring societies and non-members
- e) Market access
- f) Monitoring
- g) Leadership
- h) Overall program feedback

The members who took part in the community questionnaire usually included members of the executive committee as well as female member/s.

The household and community questionnaire are available upon request from the main author.

2. Supporting text and analysis

2.1 Forest management outcome: Potential crop trees (PCT)

The forest management outcome of each group is measured by the number of young trees per hectare. These trees are also called ‘potential crop trees’ (PCT). Here we briefly discuss the definition of PCT, how PCT outcome is calculated, and the relevance of PCT as an outcome measure.

Definition, assessment, and purpose

PCT refers to trees that are: (i) over two meters tall, (ii) have no damage to their main stem, (iii) at least 4 meters apart from each other, and (iv) depending on the species, have a diameter at breast height (DBH) of either ≤ 40 or ≤ 25 cm at maturity (15, 16).

Group	DBH	Tree Species
I	≤ 40	<i>Hagenia abyssinica</i> , <i>Podocarpus falcatus</i> , <i>Juniperus excelsa</i> , <i>Olea europea</i> , <i>Ekebergia capensis</i>
II	≤ 25	<i>Pittosporum viridiflorum</i> , <i>Hypericum lanceolatum</i> , <i>Rapania melanophloeos</i> , <i>Erica arborea</i>

PCT assessment is carried out once every five years by the forest administration with active participation from group members. The main purpose of this assessment is to determine the annual allowable timber quota and the rent each group has to pay to the local forest administration. The rent decreases in the forest outcome. The assessment is made by counting PCT in carefully sampled plots and then preparing an average for the entire forest area. The first PCT assessment was conducted in 2005; the second assessment is expected to begin in 2010. However, due to the enormous work involved in conducting and analyzing this data, the results are expected to be available sometime in 2012.

Relevance

PCT per hectare is considered a highly relevant indicator of forest management success. This is because before the launch of the forest management program, members targeted medium-sized trees for logging as these were easier to cut, process, and transport. In case of

tree species preferred for non-timber forest products (firewood, charcoal, construction and farm material), members did not wait until the trees were mature as these tree species were not used for timber production. Because of these selective practices, trees of intermediate height, such as PCT, were found to be entirely missing from the forest, resulting in a very low average annual forest increment of just 1 m³/ha/yr. Moreover, such patterns were observed across many different sites (S16, S21).

Most of the Adaba-Dodola forest is considered as degraded and the recovery potential of such a forest depends on the intensity of past extraction as well as the availability of young trees in the forest. As a result, forest management outcome based on the abundance and distribution of threatened as well as young trees with a potential to form a long-term final stand was considered important. In accordance, an inventory of the number of PCT was formally used as the criteria to evaluate the performance of a group. A tree height of two meters was taken as a benchmark on the assumption that trees above this height are beyond the reach of animals and thus can escape browsing damage. Moreover, group specific diameters were selected on the assumption that trees above this range will not have a significant impact on improving the forest quality. These features make PCT a robust measure of forest management success that covers forest quality as well as vitality (S16).

2.2 Causality

Causality is hard to prove without an ideal experiment, which in our case would have been conducted before the onset of the program. Unfortunately, this was not possible, as we did not come to know of the program before it was in place. Therefore, in this section, we provide theoretical, experimental, and statistical evidence in support of the suggested causality, which allows us to estimate the effect of conditional cooperation on the forest management outcome. Our statistical analyses is based on, inter alia, two-stage least squares regressions (2SLS) in which we use cooperativeness of the leader of a forest user group and clan affiliation as instruments for the share of conditional cooperators in the group. The instruments are based on theories of prestige-biased social learning (S22) and

possible clan differences with regard to norms of cooperation, respectively. The results from both 2SLS estimates show a positive and significant effect of conditional cooperation on forest management.

Theoretical evidence

There exist strong theoretical arguments for the suggested causality, including numerous social preference models that predict more cooperative outcomes as a function of the number of conditionally cooperative players in a group (S23-28). (S29) offer a nice summary of the theoretical explanations deriving explicit conditions based on (S25). The intuition behind all these models is that (i) conditional cooperators have different “payoffs” and thus play a different game than free riders, and (ii) the presence of conditional cooperators changes the game of free riders once sanctioning possibilities are available.

Experimental evidence

Besides the theoretical evidence, the causal relationship between conditional cooperation and cooperative outcomes has also been established by many experiments in the lab (S30-33). Typically, participants play a game similar to the one in this study at the beginning of the experiment that provides information about the behavioral type of each participant. Participants are then sorted into groups (typically, without knowing) that differ in the share of conditional cooperators. The results of the experiments clearly show that cooperation is significantly higher in groups with higher share of conditional cooperators. Moreover, they also show that a type’s behavior is stable over many interactions in a repeated social dilemma game.

Two-stage least squares regression (2SLS)

Here, we substantiate the theoretical and experimental results in support of the suggested causality with results from 2SLS. We first highlight the significantly positive effect of conditional cooperation as instrumented by ‘leader’s cooperativeness’ on the forest

management outcome. Second, we show similar results by using ‘clan affiliation’ as an alternative instrument for conditional cooperation.

a) Leader of a forest user group as an instrument

Models of cultural evolution suggest that individuals rely on various forms of social learning mechanisms to extract adaptive behaviors. These mechanisms play an important role in the transmission of behavioral traits in groups and are capable of generating, on the one hand, within group uniformity, but on the other hand, between group differences that could persist over long periods of time (S34, S35). People, however, do not imitate behaviors from individuals randomly, but are biased in whom they choose to learn from. (S22) outline that an important source of bias in transmission comes from prestige, in which individuals use a variety of cues, such as skill and age, to determine which individuals in a group are likely to have above-average information in the context in which they seek to acquire knowledge. For example, (S36) write:

“A learner should use cues, attributes, or characteristics of the individuals in their social world to figure out who is most likely to have useful ideas, beliefs, values, preferences, or strategies that might be gleaned, at least partially, through observation. (...) Finding out who possesses the more adaptive skills, strategies, preferences, and beliefs is no straightforward task. To achieve this, people rely on a range of cues related to skill (or competence), success, and prestige. (...) Cues of success are less direct and take advantage of easily observable correlates of competence. (...) Depending on the domain and society, such cues might be measured by house size, family size, number of wives and/or children (...).”

Similarly, work in social psychology has documented the important role of leadership for group cooperation. Quoting (S37):

“Two major types of motivation underlie the ability of leaders to gain cooperation in groups. First is the desire of people to gain rewards and avoid punishments. Leaders can tap into such motivations to the extent that they control resources and/or instruments of surveillance and sanctioning. Second is people’s internal attitudes and values, which shape what people want or feel they ought to do. Leaders can draw on these internal motivations by

appealing to or creating attitudes and values. Both strategies influence behavior, but there are clear advantages to leadership based on connecting to people's attitudes and values. In particular, people voluntarily follow leaders who engage their internal motivations."

The importance of a leader's cooperative type for group cooperation is also highlighted by recent experimental evidence where (S38) elicit the behavioral type of leaders in a voluntary contributions game and find that "groups perform best when led by those who are cooperatively inclined."

Leaders are prestigious in our setting

In line with these arguments, we hypothesize that freely conferred deference on the group leader could explain differences in the share of conditional cooperators between groups. Our proposition rests on the observation that leaders command a high social standing among the groups and their opinions are well respected by the group members, such that their recommendations are sought not only in the realm of forest management activities but also in wider social and economic domains. For instance, in one instance, the forest user group leader was approached to decide on the punishment of a thief responsible for stealing group members' livestock for three months. On another occasion, the saddle of a horse, which one of our field assistants was riding, was accidentally broken. The horse owner approached the leader of the group to settle the compensation that the field assistant should pay. Since these activities clearly fall beyond the purview of the duties of the leader, whose official job is restricted to activities pertaining to forest management, it seems that leaders are frequently consulted in other domains and their opinions are considered prestigious in these domains as well.

Even though, officially, leaders are elected by group members in our setting, we show below that in line with the traditional leader selection process among the Oromo people (*Gadaa*) as well as indicators of prestige, group leaders are typically older, better-educated, and richer than the other members (Table S2, S3).

This suggests that traditional criteria that can be seen as exogenous to the group play an important role in the selection of the leader. Moreover, we show that the behavioral type

of a person has no effect on the likelihood of being a leader, neither in groups with above average share of conditional cooperators nor in groups where the share is below average (Table S4). We also find no significant group effects (Table S5). Thus, we can rule out that the election of norm-conform leaders is causing the correlation between leaders and group members. Rather, it seems likely that leaders evoke prestige and their behavior is indeed preferentially copied by the group members.

Table S2 reports socio-economic and demographic measures for leaders and other group members, respectively. We find that leaders are on average older than an average member. In fact, the average age of a leader is in line with the once prominent traditional Oromo age-based class system of leader selection called *Gaada*, in which individuals in the *Luba* age grade (40-48 years) are considered responsible for decision-making. Besides age, we find that leaders also have higher education, larger family size and wealth (as measured by number of cattle, horses, and land ownership).

Our results are robust even when we split our sample into two sub-samples based on higher and lower than average share of conditional cooperators in a group and then re-run the tests (Table S3). These findings imply that irrespective of the share of conditional cooperators, leaders happen to be older, wealthier, highly educated, and with larger family size when compared to the rest of the group, indicating that traditional-prototypical representations as underlined by (S22, S36) and other ethnographic studies play an important role in leader selection.

Variable	Leaders	Members	Test Statistic
Age (years)	40.92 (1.137)	36.40 (0.585)	$z = -3.43$ $P = 0.000$
Education (years)	4.63 (0.394)	3.34 (0.131)	$z = -3.37$ $P = 0.000$
Family size	11.53 (0.719)	8.37 (0.178)	$z = -4.66$ $P = 0.000$
Number of cattle	15.54 (1.994)	11.20 (0.429)	$z = -3.26$ $P = 0.001$
Number of horses	3.16 (0.27)	2.62 (0.088)	$z = -2.14$ $P = 0.033$
Land holding (hectares)	1.89 (0.159)	1.48 (0.047)	$z = -2.96$ $P = 0.003$

Table S2. Socio-economic characteristics of leaders and group members. Figures in parenthesis are standard error of mean. The test used is Mann-Whitney.

Variable	Groups with higher than average share of conditional cooperators			Groups with lower than average share of conditional cooperators		
	Leader	Member	Test	Leader	Member	Test
Age (years)	40.32 (1.599)	36.64 (0.769)	$z = -2.12$ $P = 0.034$	41.57 (1.642)	36.15 (0.886)	$z = -2.68$ $P = 0.007$
Education (years)	4.73 (0.476)	3.58 (0.184)	$z = -2.33$ $P = 0.020$	4.52 (0.656)	3.08 (0.186)	$z = -2.39$ $P = 0.017$
Family size	11.33 (0.852)	8.66 (0.254)	$z = -3.02$ $P = 0.003$	11.73 (1.166)	8.07 (0.248)	$z = -3.64$ $P = 0.000$
Number of cattle	13.71 (1.910)	10.66 (0.498)	$z = -1.89$ $P = 0.059$	17.27 (3.455)	11.76 (0.705)	$z = -2.67$ $P = 0.008$
Number of horses	3.19 (0.418)	2.21 (0.104)	$z = -2.36$ $P = 0.019$	3.14 (0.356)	3.04 (0.138)	$z = -0.53$ $P = 0.599$
Land holding (hectares)	1.89 (0.224)	1.57 (0.071)	$z = -1.62$ $P = 0.106$	1.92 (0.230)	1.39 (0.061)	$z = -2.610$ $P = 0.009$

Table S3. Socio-economic characteristics of leaders and group members split by groups with higher and lower than average share of conditional cooperators. Figures in parenthesis are standard error of mean. The test used is Mann-Whitney.

We further test our proposition econometrically, by regressing the dummy for the person elected as a leader (1 = yes, 0 = no) on the behavioral type of the person (1 =

conditional cooperator, 0 = otherwise), controlling for age, education, family size, and wealth (Table S4). Our results show that in the sub-sample dominated by conditional cooperators, the predicted probability of a person being elected as a leader is 0.07 if the person is a conditional cooperator and 0.04 otherwise. Similarly, in the other sub-sample dominated by players other than conditional cooperators, the probabilities are 0.06 and 0.02, respectively. Table S4 shows that regardless of the sub-sample, these probabilities are not statistically different from each other. In contrast, education has a highly significant effect on the probability of being elected as a leader in both the cases.

	Dependent variable: Leader dummy (1 = leader, 0 = otherwise)	
	Groups with higher than average share of conditional cooperators	Groups with lower than average share of conditional cooperators
Conditional cooperator dummy	-0.197 (0.217)	-0.389 (0.421)
Age	0.011 (0.008)	0.014* (0.008)
Education	0.099 *** (0.032)	0.119*** (0.036)
Family size	0.055 (0.034)	0.079** (0.033)
Cattle ownership	-0.001 (0.015)	0.003 (0.009)
Constant	-2.768 (0.308)***	-3.155 (0.351)
Wald χ^2 (5)	30.14 $P = 0.000$	33.25 $P = 0.000$
Pseudo R ²	0.095	0.151
Observations	301	295

Table S4. Determinants of leader selection in groups with higher and lower than average share of conditional cooperators, respectively. Probit estimation with robust standard errors clustered on the forest user group. * $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$

Finally, we regress the dummy for the person elected as a leader (1 = yes, 0 = no) on age, education, family size, and wealth, controlling for group specific dummies (Table S5,

Model 1). The results show that though age, education and family size have a positive and significant effect, none of the group dummies are statistically significant. Consequently, dropping the dummies (Table S5, Model 2) does not affect our results. In fact, a Likelihood Ratio test shows that the group dummies do not improve the fit of the model ($P = 1.000$). Overall, our findings imply that traditional features, such as education, family size, and age, but not the behavioral type, play a crucial role in being elected as a leader.

VARIABLES	Dependent variable: Leader dummy (1 = leader, 0 = otherwise)	
	(1)	(2)
Age	0.015** (0.006)	0.012** (0.006)
Education	0.136*** (0.026)	0.105*** (0.024)
Family size	0.090*** (0.026)	0.066*** (0.024)
Cattle ownership	-0.003 (0.008)	0.001 (0.007)
Group dummies	YES	NO
Constant	-3.704*** (0.616)	-2.992*** (0.228)
Log likelihood	-127.459	-137.11
Likelihood ratio χ^2	81.10 (0.001)	49.74 (0.000)
Pseudo R ²	0.145	0.112
Observations	529	596

Table S5. Determinants of leader selection. Model 1 is Probit with robust standard errors. Model 2 is Probit with robust standard errors clustered on the group. In Model 1, 67 observations representing six groups were dropped because of perfect prediction. Though estimates in Model 2 include these observations, these were dropped to conduct the Likelihood Ratio test. * $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$

Leader's cooperation as an instrument

Having shown that leaders are indeed prestigious and that traditional criteria play an important role in the selection of the leader, here we establish the validity of our instrument, by showing while a leader's cooperativeness is correlated with the share of conditional cooperators in a group, it is not correlated with the outcome.

The behavioral type of a leader (1 = conditional cooperator, 0 = otherwise) is highly correlated with the share of conditional cooperators in a group (Mann-Whitney test, $z = -2.94$, $P = 0.003$), making it a relevant instrument. However, the relevance of our instrument is greatly increased if instead of a binary variable we use a discrete variable, like a leader's (unconditional) cooperation in the public goods experiment. The cooperation decision of a leader and his behavioral type are highly correlated (Mann-Whitney test, $z = -2.78$, $P = 0.005$). Since all our validity checks above hold for the cooperation decision as well and a leader's cooperation is also significantly correlated with the share of conditional cooperators in a group (Spearman's $\rho = 0.43$, $P = 0.002$), we use a leader's cooperation decision as an instrument for the share of conditional cooperators in that group.

For the leader to be a valid instrument we have to further ensure that leaders alone do not exert any direct effect on the forest management outcome. We don't think this assumption seems questionable, as forest management is clearly a collective action problem and one single individual alone is unlikely to have a direct effect on the forest outcome. Instead, leaders may affect the forest management outcome by influencing behavioral norms in a group to sustain group member cooperation as described above. Since group leaders were elected right at the start of the program, when the program outcome was yet unknown, it is also not possible for the outcome to have influenced the selection of the leader by group members. Finally, we believe that the leader is not subject to the same endogeneity as ordinary members, because we do not expect a leader's type to be as easily malleable as a member's type. Of course, there is no way to prove this claim explicitly with our data, but we also have no evidence that refutes this assumption.

When we test whether a group leader's type correlates with PCT, the data show that there is no difference in the forest outcome across groups with and without conditionally cooperative leaders (Mann-Whitney test, $z = -0.79$, $P = 0.429$). Similarly, we find that there is no significant correlation between the forest outcome and the unconditional contribution by leaders in the public goods game (Spearman $\rho = 0.15$, $P = 0.316$).

First and second stage regression results

First stage regression results (Table S6) highlight the relevance of our instrument. The leader's cooperativeness has a significantly positive effect on the share of conditional cooperators in a group. This result is robust even when we change the configuration of structural factors and employ different proxies to measure social and economic heterogeneity. We also interact leader's cooperativeness with time and find that though the effect of leader's cooperativeness on the share of conditional cooperators declines over time, the overall effect is positive. A post regression F test further shows that the two coefficients are significantly different from each other ($F(1, 40) = 20.39, P = 0.000$). Together, the instruments account for more than 30% of the variation in the share of conditional cooperators. A joint F test further highlights that the instruments are powerful ($F \text{ statistic} > 17$ in all models). A Hansen J statistic for over-identification shows that the instruments are valid ($P = 0.50$). Lastly, Wooldridge's robust score test and a robust regression-based test support the hypothesis that the instruments are exogenous ($P > 0.25$).

Table S7 reports the results of the second stage regression and shows that the share of conditional cooperators in a group has a significantly positive effect on the forest management outcome even when we instrument the share of conditional cooperators in a group, but the coefficients of OLS and 2SLS are nearly the same (see Table S12 for comparison). The effect is robust even when we change the configuration of the structural variables.

VARIABLES	Dependent variable: Share of conditional cooperators			
	(1)	(2)	(3)	(4)
Elevation	-0.086 (0.066)	-0.089 (0.068)	-0.094 (0.061)	-0.085 (0.065)
Group size	-0.000 (0.005)	-0.000 (0.005)	0.001 (0.005)	-0.000 (0.005)
Female share	-0.000 (0.003)			-0.000 (0.003)
Cattle heterogeneity	0.015 (0.251)	0.001 (0.268)	0.012 (0.246)	
Market distance	-0.063 (0.037)	-0.061 (0.040)	-0.052 (0.039)	-0.063 (0.038)
Time	0.057 (0.054)	0.051 (0.054)	0.050 (0.053)	0.057 (0.053)
Clan heterogeneity		-0.039 (0.143)		
Settlement heterogeneity			-0.158 (0.130)	
Land heterogeneity				0.004 (0.272)
Leader's cooperativeness	0.136*** (0.024)	0.131*** (0.027)	0.139*** (0.024)	0.136*** (0.024)
Leader's cooperativeness*time	-0.109*** (0.033)	-0.102** (0.043)	-0.111*** (0.031)	-0.109*** (0.033)
Constant	0.253 (0.172)	0.274 (0.165)	0.288 (0.185)	0.257 (0.186)
R-squared	0.441	0.442	0.457	0.441
Adj.R-squared	0.329	0.331	0.348	0.329
First Stage Regression F statistic	18.01	17.73	17.95	17.49
Robust score $\chi^2(1)$	0.80 $P = 0.371$	0.39 $P = 0.531$	1.31 $P = 0.253$	0.62 $P = 0.431$
Robust regression F (1, 40)	0.67 $P = 0.418$	0.32 $P = 0.572$	1.15 $P = 0.291$	0.49 $P = 0.487$
Hansen J statistics $\chi^2(1)$	0.20 $P = 0.657$	0.15 $P = 0.699$	0.39 $P = 0.534$	0.44 $P = 0.507$
Observations	49	49	49	49

Table S6. Determinants of conditional cooperator shares in a group. The models reported here are first stage reduced form regressions. Models 1 to 3 vary in social heterogeneity controls: Model 1 controls for share of female members, model 2 for clan heterogeneity, and model 3 for settlement heterogeneity. Model 1 and 4 vary in how economic heterogeneity is measured. Model 1 measures economic heterogeneity as cattle heterogeneity, model 4 measures it as heterogeneity in land ownership. ** $P < 0.05$, *** $P < 0.01$

VARIABLES	Dependent variable: Forest management outcome			
	1	2	3	4
Conditional cooperator share	66.564*** (21.964)	59.959** (23.897)	68.452*** (21.710)	64.807*** (24.589)
Elevation	20.641*** (7.072)	19.593*** (7.272)	20.492*** (7.052)	17.873** (7.643)
Group size	0.471 (0.513)	0.528 (0.535)	0.585 (0.534)	0.554 (0.554)
Female share	0.160 (0.190)			0.199 (0.198)
Cattle heterogeneity	-58.760* (34.927)	-62.166* (36.111)	-61.313* (34.351)	
Market distance	-19.954*** (3.937)	-19.765*** (3.783)	-18.561*** (4.180)	-19.477*** (4.091)
Time	30.991*** (5.700)	30.972*** (4.863)	31.268*** (5.369)	31.477*** (6.188)
Clan heterogeneity		-11.271 (14.089)		
Settlement heterogeneity			-15.981 (18.116)	
Land heterogeneity				-28.177 (31.860)
Constant	74.798*** (21.527)	84.267*** (22.464)	80.701*** (21.563)	60.365** (24.078)
R-squared	0.711	0.718	0.713	0.698
Adj.R-squared	0.662	0.670	0.664	0.646
Observations	49	49	49	49

Table S7. Determinants of forest management outcome. 2SLS with robust standard errors. Models 1 to 3 vary in social heterogeneity controls: Model 1 controls for the share of female members, model 2 for clan heterogeneity, and model 3 for settlement heterogeneity. Model 4 takes heterogeneity in land ownership as a measure of economic heterogeneity. In all four models, share of conditional cooperators in a group is a significant predictor of the forest management outcome. * $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$

b) Alternative instrument: Clan affiliation

One potential concern regarding the leader as an instrument mentioned already above might be that the type of the leader is not without doubt exogenous. At best, one would like to have a piece of information about forest users that is clearly exogenous to the program, has no connection to forest management and yet affects users' attitudes towards cooperation. Clan affiliation has exactly these features. The results we generate using clan affiliation as an instrument for conditional cooperation replicate our findings from the previous analysis with leader's cooperativeness and provide further strong support in our interpretation of the direction of causality. In particular, because clan affiliation is clearly exogenous the results are hard to explain by reverse causality.

The Oromo society is sub-divided into many clans, which are based on kinship, tracked by lineal descent from a common ancestor. We conjecture based on limited ethnographic accounts on the Oromo society (S39, S40) that clans may actually differ in their social norms and institutions leading to different attitudes towards cooperation and hence shares of conditional cooperators. In line with this hypothesis we find that on an individual level, there is a negative correlation between a person belonging to one of the major clans, henceforth called 'clan A', and the person being a conditional cooperator ($\chi^2(1) = 5.03, P = 0.025$). Similarly, at the group level, we find that groups with members from this clan have lower share of conditional cooperators (Mann-Whitney test, $z = 2.919, P = 0.004$). Because a member is born into a clan, it is clearly exogenous and thus not subject to reverse causality. In addition, there is no evidence that forest users from this particular clan are different from users from other clans with regard to any aspect pertaining to forest management. This is important to ensure that clan affiliation has no direct effect on forest outcome. For example, users do not differ with regard to age (Mann-Whitney test, $z = -0.19, P = 0.847$), family size (Mann-Whitney test, $z = 0.18, P = 0.856$), cattle ownership (Mann-Whitney test, $z = -1.31, P = 0.190$), financial status ($\chi^2(2) = 3.73, P = 0.155$), or the share of income they derive from forest products (Mann-Whitney test, $z = -0.258, P = 0.796$). The only avenues where we find a small yet significant difference is education (Mann-Whitney test, $z = -4.07, P = 0.000$) and land ownership (Mann-Whitney test, $z = -1.95, P = 0.051$), with members from clan A having on average only 1.4 years less

education and 0.18 hectares more land. As Oromo is a pastoral community where land is neither the main asset nor the main source of income, none of these two variables can have a plausible impact on forest management. Further, results from an econometric analysis (Table S8) confirm that clan affiliation has no direct effect on forest outcome ($P = 0.648$). This allows us to use the presence and absence of members from clan A in the group (0 = members from clan A absent, 1 = members from clan A present) as a valid instrument for conditional cooperation.

VARIABLES	Dependent variable: Forest management outcome
Conditional cooperator share	49.60*** (16.068)
Clan A	-4.082 (8.884)
Elevation	19.097** (7.291)
Group size	0.414 (0.533)
Female share	0.135 (0.215)
Cattle heterogeneity	-55.660 (38.876)
Market distance	-20.280*** (4.144)
Time	29.980*** (7.347)
Constant	84.521*** (20.711)
R-squared	0.719
Adj. R-squared	0.663
Observations	49

Table S8. Forest management outcome, conditional cooperation, and clan affiliation. OLS with robust standard errors in parentheses. ** $P < 0.05$, *** $P < 0.01$

Table S9 reports the results from the first stage regression. We find that the presence of clan A has a significantly negative effect on the share of conditional cooperators in a group, even when we control for other covariates. In Table S10, we report the results from the 2SLS, where we instrument conditional cooperation with the clan dummy. Our results show that conditional cooperation has a significantly positive effect on PCT. Moreover, the coefficient of conditional cooperation is strikingly similar to the one obtained in IV estimates with leader's cooperativeness as an instrument. The same holds for all other covariates, as well as for R-squared and adjusted R-squared (cf. Table S7).

VARIABLES	Dependent variable: Share of conditional cooperators
Clan A	-0.219** (0.102)
Elevation	-0.091 (0.072)
Group size	-0.003 (0.006)
Female share	-0.001 (0.003)
Cattle heterogeneity	0.184 (0.309)
Market distance	-0.021 (0.039)
Time	-0.047 (0.096)
Constant	0.563** (0.209)
R-squared	0.264
Adj. R-squared	0.139
Observations	49

Table S9. Determinants of conditional cooperator shares in a group. The model reported is first-stage reduced form regression. ** $p < 0.05$

VARIABLES	Dependent variable: Forest management outcome
Conditional cooperator share	68.207** (31.171)
Elevation	20.798*** (7.211)
Group size	0.476 (0.516)
Female share	0.161 (0.186)
Cattle heterogeneity	-59.076* (34.361)
Market distance	-19.884*** (3.944)
Time	30.856*** (6.258)
Constant	74.044*** (23.545)
R-squared	0.709
Adj. R-squared	0.660
Observations	49

Table S10. Determinants of forest management outcome. 2SLS with robust standard errors in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Further test results

Finally, we report the results from several additional tests to analyze if our data provide evidence in support of reverse causality. The hypothesis underlying reverse causality is that cooperative types are not exogenous but are the result of successful forest management.

If reverse causality holds and groups learn over time to become cooperative, then we should observe higher share of conditional cooperators in groups that were established earlier. However, this is not the case (Mann-Whitney test, $z = -1.43$, $P = 0.153$). Further econometric analyses shows that the coefficient of the interaction between conditional cooperation and time is not statistically different from zero ($P = 0.78$). We thus cannot conclude that older groups have become more cooperative over time or the effect of conditional cooperation on forest commons management is mediated by time.

Moreover, if the hypothesis was correct, groups established earlier should also be more homogenous with regard to group members' behavioral type. Again, this is not what we find. To analyze a group's behavioral heterogeneity, we use for each group member the Spearman correlation coefficient between self and partner player's contribution in the conditional decision of the public goods game. We call this variable the 'cc coefficient'. One possibility to measure heterogeneity in the type composition of a group is to take the coefficient of variation (CV) of this variable. An alternative is the variance to mean ratio (VMR). Irrespective of the measure, we do not find any difference with regard to behavioral heterogeneity between groups established earlier and groups that were established later (Mann-Whitney test; CV: $z = -0.24$, $P = 0.810$; VMR: $z = -0.34$, $P = 0.734$). Thus, forest users who have had more time to interact with each other do not display more homogeneity, different to what reverse causality would suggest.

Finally, we test whether groups whose members are on average younger exhibit a stronger relationship between conditional cooperation and forest management outcome than groups whose members are on average older. The intuition is that if reverse causality holds, it should be stronger for groups with younger members, as their preferences are typically more malleable. We test this econometrically, by interacting conditional cooperation with the variable 'average age of members in a group'. The results show again that the interaction term is not statistically different from zero ($P = 0.420$), implying that the relationship between conditional cooperation and PCT does not differ between groups with younger and older members.

Taken together, we do not find any empirical evidence in our data that provide support for reverse causality. At the same time, all results are consistent with the opposite assumption that the behavioral type composition of a group affects the outcome of forest management.

2.3 Structural factors

In this section, we present an overview of the structural factors used in the regression models. We include variables commonly found to be of relevance in studies on commons management (S41-43): resource characteristics, group size, social heterogeneity, economic

heterogeneity including wealth, and market distance. In addition, we also control for time when a group was formally established, relations with neighboring groups, and demographic factors (age, education, family size). Table S11 lists proxies we apply for each of these controls. Given our sample size, we use only one proxy per control at a time to allow for reasonable degrees of freedom.

Elevation

Adaba and Dodola forests lie on the north face of the Bale Mountains, separated by a distance of roughly 40 km. Due to this relatively small distance, there seems to be little variation, if any, in slope, precipitation, soil, and temperature patterns between the two forests. However, variation in these factors is more pronounced across elevation, which ranges roughly from 2600 to 3500 m. Because climatic factors, such as slope, temperature, soil, and rainfall are correlated with each other, we expect elevation to be an important factor in mediating their effect on forest growth. Moreover, higher elevation is also likely to deter humans and livestock from accessing the forest (S44). Thus, elevation is likely to have a positive effect on the forest management outcome.

Because forests blocks managed by the groups span a range of altitudes, it is difficult to have a precise measure on elevation. Here, we use secondary data on the presence and absence of Erica heather as an indicator of elevation. In the Bale Mountains, Erica heather grows upwards of 3000 meters. In accordance, we treat elevation as a dummy variable (0 = Erica forest absent and thus lower elevation, 1 = Erica forest present and thus higher elevation). 12 groups have forest blocks at higher elevation and all of them are relatively inaccessible.

Group size

Under the participatory forest management program, each group is allowed to have a maximum of 30 members. Actual size among the 49 groups varies from 16 to 30 members, with 40 % of the groups having 30 members. A variety of relationships between group size and cooperation are posited in the literature; these range from negative and positive to curvilinear (S43).

Social heterogeneity

Groups vary socially in their share of female members, clan composition, and number of settlement clusters. These are briefly discussed below:

Share of female members: Groups differ in their share of female members. Such differences could affect forest management outcomes because women play an important role in collecting forest products but are often relegated to a marginal position in decision-making (S43). In the Bale Mountains, women are involved in activities, such as gathering forest products and grazing livestock, which have a substantial effect on forest growth. However, because women play only a marginal role in the Oromo society, involving them in the decision-making process is likely to have a positive effect on the forest management outcome.

Settlement heterogeneity: Most groups are organized into small hamlets, which may or may not lie adjacent to each other. It is hypothesized that settlement heterogeneity may create obstacles to mutual understanding and trust, resulting in a negative effect on the forest management outcome. We follow (45) and use an index of fractionalization to measure settlement heterogeneity, given by $1 - \sum (P_i)^2$, where P_i denotes the proportion of total population in the i^{th} settlement. The index varies from 0 to 1 and measures the probability that two persons selected randomly from a group will not be from the same settlement.

Clan heterogeneity: Groups also differ in clan composition. While some are homogeneous, others have members from different clans. Just as with settlement heterogeneity, it is hypothesized that clan heterogeneity may create obstacles to mutual understanding and trust. Clan heterogeneity is measured analogously to settlement heterogeneity.

Economic variables: wealth and wealth inequality

Wealth and wealth inequality are considered important factors affecting the success of commons management. In an experimental study conducted with commons users, (S46) report that cooperation outcome in the experiment was lower in groups with higher average wealth as well as higher variation in the distribution of wealth. Others, however posit from

a positive to a U-shaped relationship between wealth inequality and cooperation (S42, S43, S45). In a herding society like ours, livestock rearing, particularly cattle ownership is considered an important economic activity and serves as an indicator of the wealth status of an individual. A secondary indicator of wealth is land, which is mainly used for subsistence farming. We aggregate individual level data on cattle and land ownership to measure average cattle and land wealth of a group. Because wealth inequality in our setting is due to inter-individual differences in cattle and land ownership, we use the Gini index of cattle and land ownership to measure these, respectively. Due to our sample size ($N = 49$), we do not consider higher order terms.

Market distance

In many groups, local rules and regulations on forest use allow members to sell forest products in weekly markets. Profitability of this, however, depends on the distance of a group from the market. Access to markets is much better in groups located at lower elevations and near the main towns, such as Adaba and Dodola. By contrast, for groups at higher elevations, it takes members nearly four hours to reach the nearest market. This has strong consequences on the extent to which members can earn cash income by selling forest products in the market. Even though market access varies by elevation (Mann-Whitney $z = -2.83$, $P = 0.005$), we do not detect multicollinearity in any of our models. The effect of market access on cooperation is disputed in the literature, with some authors suggesting that better market access induces members to take higher interest in forest management activities, while others suggest that markets provide people with exit options leading to weakening of this interest (S42, S43). In our specific field settings, we expect market access to have a positive effect on the forest management outcome. We measure market distance as the average time (number of hours) it takes for a group to reach the nearest market.

Time

Groups in the forest management program were not established at the same point in time but in two phases over a period of five years. We expect this to have an influence on the

forest management outcome through tree growth. Therefore, time is expected to have a positive effect on the forest management outcome. Because we do not have precise data on the date of establishment of each group, we model time as a dummy variable (0 = groups established between 2003 and 2005, and 1 = groups established between 2000 and 2002).

Relation with neighboring groups

Forest user groups are either surrounded by other such groups or non-PFM groups. Depending on the relationship a group has with its neighbors, the presence of neighboring groups might have an effect on the forest management outcome. To check for this, we asked each group in the community level survey to rate their relationship with the neighboring (if any) forest user and non-PFM groups on a three point scale: good, normal, and bad. We find that of the 63 neighboring non-PFM groups, forest user groups reported good relations with 26, normal with 29, and bad with just 8 groups. The results with respect to neighboring forest user groups are as follows: for a total of 171 neighboring forest user groups, an overwhelming rated good relationship with 120 groups, normal with 29 groups and bad with 22 groups. Results from econometric analyses further show that sharing border with another forest user groups ($P = 0.493$) or with non-PFM groups ($P = 0.906$) has no effect on the commons outcome. Given that most forest user groups have normal to good relations with their neighbors, these results are not surprising.

Demographic factors

Groups vary substantially in demographic factors such as age, education, and family size, which might also influence the forest management outcome of a group. We measure these by aggregating individual responses in the household questionnaire at the level of a group. Thus, in the regression models we control for average age in years, average number of years of education, and average family size.

Structural factor	Proxy variable	Mean	SE	Minimum	Maximum
Resource	Elevation	0.25	0.062	0	1
Group	Group size	26.16	0.665	16	30
Social heterogeneity	Share of female members	20.19	1.657	0	66.67
	Settlement heterogeneity	0.40	0.034	0	0.78
	Clan heterogeneity	0.55	0.026	0	0.76
Economic variables	Cattle	11.44	0.537	5.85	23.58
	Land	1.49	0.073	0.35	3.00
	Cattle heterogeneity	0.34	0.012	0.19	0.59
	Land heterogeneity	0.30	0.011	0.02	0.43
Market distance	Market distance	2.37	0.125	0.75	4.5
Time	Time	0.51	0.072	0	1
Neighboring groups	Forest user groups	3.57	0.210	1	8
	Non-PFM groups	1.29	0.16	0	5
Demography	Age	36.96	0.882	26.00	57.13
	Education	3.42	0.162	1.13	5.92
	Family size	8.61	0.242	5.64	13.44

Table S11. Summary statistics on structural variables. SE = standard error

2.4 Supplementary regression models

In this section, we first present Model 1 and 2 listed in the main paper but with a complete listing of the control factors (Table S12). Second, we present a series of supplementary regression models, wherein we show that the effect of conditional cooperators on the forest management outcome is robust even when we (i) use a weighted least squares regression to account for differences in sample size across groups (Table S13), (ii) change the configuration of control factors and use different proxies to measure social (Table S14) and economic characteristics (Table S15), (iii) control for demographic factors as age, education and family size (Table S16), and (iv) measure conditional cooperation by

combining the share of both strong and weak conditional cooperators (Table S17). Third, we obtain theoretically expected and robust results when we categorize groups by their share of free riders. As expected, free rider shares have a negative effect on the forest management outcome (Table S18).

VARIABLES	Dependent variable: Forest management outcome	
	1	2
Conditional cooperator share		52.085*** (13.117)
Elevation	14.280* (7.970)	19.257** (7.215)
Group size	0.260 (0.590)	0.425 (0.525)
Female share	0.116 (0.263)	0.151 (0.203)
Cattle heterogeneity	-45.924 (41.598)	-55.968 (37.937)
Market distance	-22.779*** (4.355)	-20.568*** (4.057)
Time	36.479*** (6.409)	32.185*** (6.004)
Constant	105.325*** (21.719)	81.438*** (19.186)
Adj. R-squared	0.579	0.669
Observations	49	49

Table S12. Forest management outcome, structural factors, and conditional cooperation. Model 1 shows the effect of structural factors alone on the forest management outcome. Model 2 shows the effect of conditional cooperation, controlling for structural factors. * $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$

VARIABLES	Dependent variable: Forest management outcome
Conditional cooperator share	48.950*** (13.716)
Elevation	18.997** (7.165)
Group size	0.306 (0.565)
Female share	0.120 (0.226)
Cattle heterogeneity	-47.256 (39.006)
Market distance	-20.525*** (4.095)
Time	33.286*** (6.160)
Constant	81.993*** (19.795)
R-squared	0.701
Adj. R-squared	0.650
Observations	49

Table S13. Forest management outcome, structural factors and conditional cooperation. Observation are weighted by their sample size and the estimation method is weighted least squares with robust standard errors. ** $P < 0.05$, *** $P < 0.01$

VARIABLES	Dependent variable: Forest management outcome	
	1	2
Conditional cooperator share	50.570*** (12.991)	48.973*** (13.735)
Elevation	18.672** (7.129)	18.382** (7.020)
Group size	0.534 (0.552)	0.500 (0.557)
Settlement heterogeneity	-17.582 (20.977)	
Clan heterogeneity		-13.128 (14.623)
Livestock heterogeneity	-57.731 (37.231)	-60.404 (39.528)
Market distance	-19.215*** (4.353)	-20.185*** (3.876)
Time	32.628*** (5.721)	31.599*** (5.015)
Constant	89.186*** (18.939)	89.904*** (20.305)
R-squared	0.723	0.722
Adj. R-squared	0.675	0.675
Observations	49	49

Table S14. Forest management outcome, conditional cooperation, and social heterogeneity. Model 1 and 2 vary in how social heterogeneity is represented. In Model 1, social heterogeneity is measured as settlement heterogeneity, in Model 2 it is measured as clan heterogeneity. Estimation method is OLS with robust standard errors. ** $P < 0.05$, *** $P < 0.01$

VARIABLES	Dependent variable: Forest management outcome		
	1	2	3
Conditional cooperator share	52.861*** (14.218)	49.066*** (14.085)	51.284*** (14.435)
Elevation	21.408** (8.328)	18.615** (8.125)	16.742** (7.878)
Group size	0.496 (0.546)	0.402 (0.623)	0.507 (0.567)
Female share	0.155 (0.204)	0.237 (0.226)	0.188 (0.216)
Cattle wealth	-0.912 (0.818)		
Land wealth		8.389 (6.208)	
Land heterogeneity			-24.110 (33.395)
Market distance	-20.635*** (4.159)	-19.336*** (4.338)	-20.114*** (4.167)
Time	31.476*** (5.995)	34.418*** (6.004)	32.534*** (6.376)
Constant	70.911*** (17.848)	46.158** (20.751)	66.543*** (21.032)
R-squared	0.708	0.712	0.704
Adj. R-squared	0.658	0.663	0.653
Observations	49	49	49

Table S15. Forest management outcome, conditional cooperation, and economic heterogeneity. Model 1, 2, and 3 vary in how wealth and wealth inequality are measured. In Model 1, wealth is measured as average cattle ownership, in Model 2 it is measured as average land ownership. In Model 3, heterogeneity in land ownership is used as a proxy for economic heterogeneity. Estimation method is OLS with robust standard errors. ** $P < 0.05$, *** $P < 0.01$

VARIABLES	Dependent variable: Forest management outcome
Conditional cooperator share	60.297*** (13.549)
Age	0.775 (0.714)
Education	-3.590 (3.588)
Family size	-1.560 (1.777)
Elevation	19.882** (8.007)
Group size	0.808 (0.527)
Females share	0.222 (0.216)
Cattle heterogeneity	-51.193 (43.280)
Market access	-19.612*** (4.585)
Time	35.907*** (6.279)
Constant	58.300 (39.438)
R-squared	0.746
Adj. R-squared	0.679
Observations	49

Table S16. Forest management outcome, conditional cooperation, and demographic factors. Estimation method is OLS with robust standard errors. ** $P < 0.05$, *** $P < 0.01$

VARIABLES	Dependent variable: Forest management outcome
Weak and strong conditional cooperator share	45.153*** (13.794)
Elevation	21.882** (8.157)
Group size	0.463 (0.527)
Females share	0.202 (0.216)
Cattle heterogeneity	-56.424 (35.821)
Market access	-21.419*** (4.319)
Time	32.174*** (6.385)
Constant	77.954*** (20.730)
R-squared	0.710
Adj. R-squared	0.660
Observations	49

Table S17. Forest management outcome and combined share of conditional cooperators and weak conditional cooperator. OLS estimates with robust standard errors. ** $P < 0.05$, *** $P < 0.01$

VARIABLES	Dependent variable: Forest management outcome
Free rider share	-67.100*** (17.080)
Elevation	25.205*** (8.666)
Group size	0.161 (0.561)
Female share	0.217 (0.222)
Cattle heterogeneity	-46.770 (33.399)
Market access	-24.680*** (3.678)
Time	36.660*** (5.869)
Constant	115.674*** (22.279)
R-squared	0.724
Adj. R-squared	0.676
Observations	49

Table S18. Forest management outcome and free rider shares. OLS estimates with robust standard errors in parenthesis. *** $P < 0.01$

2.5 Costly monitoring

We use community and household questionnaires to collect data on time groups and individuals spend on monitoring their forest, respectively. In the community questionnaire, we asked 5-6 members from a group about the average number of days per month and hours per day a group member spends on monitoring the forest. In the household questionnaire, we asked individual members (not interviewed earlier) the same question to obtain monitoring estimates at the level of an individual. Collecting data at two levels allows us first, to cross-check the consistency of our monitoring estimate and second, to prove the robustness of our regression results with regard to monitoring reported in the main paper.

We find that at an individual level, conditional cooperators monitor the most, spending on average 32 hours per month, 1.5 times more than free riders, who spend on average 21 hours per month. At the group level, we find that our monitoring estimates are highly consistent with those obtained at the individual level. This is evident from the highly positive and significant correlation between the estimates obtained from the community questionnaire and those obtained by aggregating individual responses in the household questionnaire (Spearman's $\rho = 0.87$, $P = 0.000$). Further, the effect of conditional cooperation on monitoring is positive and highly significant, irrespective of whether we use monitoring estimates from community (Model 2, Table S19) or household questionnaires (Model 2, Table S20) as the dependent variable. These results imply that groups with higher conditional cooperator shares spent more time on monitoring their forest. In contrast, a higher share of free riders has a significantly negative effect on monitoring (Model 3, Table S19, S20). Among the control factors, while time has a significantly positive effect on monitoring, market distance has a significantly negative effect. In the econometric models, we control for monitoring difficulty, which is a dummy variable (0 = monitoring is moderately or not difficult at all, and 1 = monitoring is very difficult or difficult). The data on monitoring difficulty was collected through the community questionnaire in which group members were asked to rate how difficult it is for them to monitor their forest. However, the effect of monitoring difficulty is not significantly different from zero in all the models.

VARIABLES	Dependent variable: Time spent monitoring in hours (community response)		
	1	2	3
Conditional cooperator share		23.782*** (8.700)	
Free rider share			-21.195** (8.543)
Monitoring difficulty	2.377 (3.808)	0.117 (3.598)	1.891 (3.476)
Group size	-0.569 (0.429)	-0.503 (0.377)	-0.531 (0.427)
Female share	0.113 (0.131)	0.113 (0.106)	0.155 (0.116)
Cattle heterogeneity	-6.156 (23.492)	-13.539 (22.072)	-5.156 (25.876)
Market distance	-5.972*** (1.518)	-4.951*** (1.543)	-5.824*** (1.630)
Time	11.428*** (3.814)	9.079** (3.477)	11.857*** (3.634)
Constant	49.467*** (16.464)	42.092** (15.553)	49.436*** (17.824)
R-squared	0.407	0.509	0.474
Adj. R-squared	0.313	0.416	0.375
Observations	45	45	45

Table S19. Determinants of average time-spent by a group on monitoring (as measured in the community questionnaire). OLS estimates with robust standard errors in parentheses. ** $P < 0.05$, *** $P < 0.01$.

VARIABLES	Dependent variable: Time spent monitoring in hours (Individual response aggregated at group level)		
	1	2	3
Conditional cooperator share		27.575*** (7.092)	
Free rider share			-21.585*** (6.526)
Monitoring difficulty	1.784 (3.390)	0.364 (2.925)	1.431 (3.162)
Group size	-0.741* (0.386)	-0.604* (0.314)	-0.701* (0.370)
Female share	0.011 (0.161)	0.035 (0.117)	0.054 (0.134)
Cattle heterogeneity	-7.394 (17.701)	-11.485 (16.420)	-5.295 (17.954)
Market distance	-6.296*** (1.476)	-4.720*** (1.155)	-6.157*** (1.463)
Time	10.867*** (3.091)	8.833*** (2.686)	11.218*** (2.927)
Constant	57.963*** (13.185)	43.567*** (11.828)	57.468*** (13.470)
R-squared	0.452	0.618	0.525
Adj. R-squared	0.374	0.552	0.444
Observations	49	49	49

Table S20. Determinants of average time-spent by a group on monitoring (as measured by aggregating individual responses in the household questionnaire). OLS with robust standard errors in parentheses. * $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$.

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